

PF Lab 8 tasks (7)

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Task 4

```
#include<stdio.h>

int main()
{
    int arr[3][3];

    printf("Enter the values for the array:\n");

    for(int i=0;i<3;i++){
        for(int j=0;j<3;j++){
            printf("Element %d%d: ",i,j);
            scanf("%d",&arr[i][j]);
        }
    }

    printf("The matrice is:\n");

    for(int i=0;i<3;i++){
        for(int j=0;j<3;j++){
            printf("%d ",arr[i][j]);
        }
        printf("\n");
    }

    if(arr[0][1]==0 && arr[0][2]==0 && arr[1][2]==0){
        printf("The matrice is lower triangular\n");
    }

    else if(arr[1][0]==0 && arr[2][0]==0 && arr[2][1]==0){
```

```

        printf("The matrice is upper triangular\n");
    }

    else{

        printf("Neither\n");

    }

    return 0;

}

```

```

C:\Users\Anas Ali Khan\Documents\PF Lab8 task4.exe
Enter the values for the array:
Element 00: 1
Element 01: 0
Element 02: 0
Element 10: 5
Element 11: 3
Element 12: 0
Element 20: 9
Element 21: 7
Element 22: 1
The matrice is:
1 0 0
5 3 0
9 7 1
The matrice is lower triangular
-----
Process exited after 18.97 seconds with return value 0
Press any key to continue . . .

```

```

C:\Users\Anas Ali Khan\Documents\PF Lab8 task4.exe
Enter the values for the array:
Element 00: 1
Element 01: 2
Element 02: 3
Element 10: 0
Element 11: 4
Element 12: 5
Element 20: 0
Element 21: 0
Element 22: 8
The matrice is:
1 2 3
0 4 5
0 0 8
The matrice is upper triangular
-----
Process exited after 15.67 seconds with return value 0
Press any key to continue . . .

```

Task 5

```
#include <stdio.h>
```

```
int main() {
```

```
    int r;
```

```
    printf("Enter number of rows: ");
```

```
    scanf("%d", &r);
```

```
    for(int i=1;i<=r;i++){
```

```
        for(int k=1;k<=r-i;k++) {
```

```
            printf(" ");
```

```
        }
```

```
        for(int j=1;j<=i;j++) {
```

```
            printf("* ");
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
    for(int i=r-1;i>=1;i--) {
```

```
        for(int k=1;k<=r-i;k++) {
```

```
            printf(" ");
```

```

    }

    for(int j=1;j<=i;j++) {

        printf("* ");

    }

    printf("\n");

}

return 0;

}

```

```

C:\Users\Anas Ali Khan\Documents\PF Lab8 task5.exe
Enter number of rows: 5
    *
   * *
  * * *
 * * * *
* * * * *
 * * * *
  * * *
   * *
    *

-----
Process exited after 1.68 seconds with return value 0
Press any key to continue . . .

```

Task 6

```

#include<stdio.h>

int main()

{

    int arr1[3][3];

    int arr2[3][3];

    for(int i=0;i<3;i++){

        for(int j=0;j<3;j++){

```

```
        printf("Enter the value for element %d%d: ",i,j);

        scanf("%d",&arr1[i][j]);

    }

}
```

```
printf("\nOriginal Matrice\n");

for(int i=0;i<3;i++){

    for(int j=0;j<3;j++){

        printf("%d ",arr1[i][j]);

    }

    printf("\n");

}

printf("\n\n");

for(int i=0;i<3;i++){

    for(int j=0;j<3;j++){

        arr2[i][j]=arr1[j][i];

    }

}
```

```
printf("Transposed Matrice\n");

for(int i=0;i<3;i++){

    for(int j=0;j<3;j++){

        printf("%d ",arr2[i][j]);

    }

}
```

```

        printf("\n");
    }

    return 0;
}

```

```

C:\Users\Anas Ali Khan\Documents\PF Lab8 task6.exe
Enter the value for element 00: 1
Enter the value for element 01: 2
Enter the value for element 02: 3
Enter the value for element 10: 4
Enter the value for element 11: 5
Enter the value for element 12: 6
Enter the value for element 20: 7
Enter the value for element 21: 8
Enter the value for element 22: 9

Original Matrice
1 2 3
4 5 6
7 8 9

Transposed Matrice
1 4 7
2 5 8
3 6 9

-----
Process exited after 6.842 seconds with return value 0
Press any key to continue . . .

```

Task 7

```

#include<stdio.h>

int main()
{
    int mat1[2][2], mat2[2][2];

    int product[2][2];

    printf("Enter values for first matrice:\n");

    for(int i=0;i<2;i++){
        for(int j=0;j<2;j++){

```

```
        printf("Element %d%d: ",i,j);  
        scanf("%d",&mat1[i][j]);  
    }  
}
```

```
printf("The first matrice is:\n");  
for(int i=0;i<2;i++){  
    for(int j=0;j<2;j++){  
        printf("%d ",mat1[i][j]);  
    }  
    printf("\n");  
}
```

```
printf("\nEnter values for second matrice:\n");  
for(int i=0;i<2;i++){  
    for(int j=0;j<2;j++){  
        printf("Element %d%d: ",i,j);  
        scanf("%d",&mat2[i][j]);  
    }  
}
```

```
printf("The second matrice is:\n");  
for(int i=0;i<2;i++){  
    for(int j=0;j<2;j++){  
        printf("%d ",mat2[i][j]);  
    }  
}
```



```

        }

        printf("\n");
    }

    printf("\n\n");

    for(int i=0;i<2;i++){
    for(int j=0;j<2;j++){

        product[i][j] = mat1[i][j] * mat2[i][j];
    }
}

printf("The product matrice is:\n");

for(int i=0;i<2;i++){

    for(int j=0;j<2;j++){

        printf("%d ",product[i][j]);

    }

    printf("\n");

}

return 0;

}

```

```
C:\Users\Anas Ali Khan\Documents\PF Lab8 task7.exe
Enter values for first matrice:
Element 00: 2
Element 01: 4
Element 10: 6
Element 11: 8
The first matrice is:
2 4
6 8

Enter values for second matrice:
Element 00: 1
Element 01: 3
Element 10: 5
Element 11: 7
The second matrice is:
1 3
5 7

The product matrice is:
2 12
30 56

-----
Process exited after 13.33 seconds with return value 0
```

Task 8

```
#include<stdio.h>

int main()
{
    int arr[3][3];

    int e=0,o=0,p=0,n=0,zero=0;

    printf("Enter the values for the matrice:\n");

    for(int i=0;i<3;i++){
        for(int j=0;j<3;j++){
            printf("Element %d%d: ",i,j);
            scanf("%d",&arr[i][j]);
        }
    }

    printf("\nThe matrice is:\n");
```

```
for(int i=0;i<3;i++){  
    for(int j=0;j<3;j++){  
        printf("%d ",arr[i][j]);  
    }  
    printf("\n");  
}
```

```
for(int i=0;i<3;i++){  
    for(int j=0;j<3;j++){  
        if(arr[i][j]%2==0){  
            e+=1;  
        }  
        else{  
            o+=1;  
        }  
  
        if(arr[i][j]>0){  
            p+=1;  
        }  
        else if(arr[i][j]<0){  
            n+=1;  
        }  
        else{  
            zero+=1;  
        }  
    }  
}
```

```

    }

}

printf("\nNumber of even values:%d\n",e);

printf("Number of odd values:%d\n",o);

printf("Number of positive values:%d\n",p);

printf("Number of negative values:%d\n",n);

printf("Number of values equals to zero:%d\n",zero);


return 0;

}

```

```

C:\Users\Anas Ali Khan\Documents\PF Lab8 task8.exe
Enter the values for the matrix:
Element 00: -2
Element 01: 0
Element 02: 9
Element 10: 4
Element 11: -7
Element 12: 3
Element 20: 0
Element 21: -1
Element 22: 5

The matrix is:
-2 0 9
4 -7 3
0 -1 5

Number of even values:4
Number of odd values:5
Number of positive values:4
Number of negative values:3
Number of values equals to zero:2

-----
Process exited after 45.71 seconds with return value 0
Press any key to continue . . .

```

Task 9

```
#include<stdio.h>

int main()
{
    int arr1[3][3];

    int arr2[3][3];

    for(int i=0;i<3;i++){
        for(int j=0;j<3;j++){
            printf("Enter the value for element %d%d: ",i,j);

            scanf("%d",&arr1[i][j]);
        }
    }

    printf("\nOriginal Matrice\n");

    for(int i=0;i<3;i++){
        for(int j=0;j<3;j++){
            printf("%d ",arr1[i][j]);
        }

        printf("\n");
    }

    printf("\n\n");

    for(int i=0;i<3;i++){
        for(int j=0;j<3;j++){
            arr2[i][j]=arr1[j][i];
        }
    }
```

```
}
```

```
int temp;
```

```
for(int i=0;i<3;i++){
```

```
    int temp=arr2[i][0];
```

```
    arr2[i][0]=arr2[i][2];
```

```
    arr2[i][2]=temp;
```

```
}
```

```
printf("Rotated Matrice\n");
```

```
for(int i=0;i<3;i++){
```

```
    for(int j=0;j<3;j++){
```

```
        printf("%d ",arr2[i][j]);
```

```
    }
```

```
    printf("\n");
```

```
}
```

```
return 0;
```

```
}
```

```
C:\Users\Anas Ali Khan\Documents\PF lab8 task9.exe
Enter the value for element 00: 1
Enter the value for element 01: 2
Enter the value for element 02: 3
Enter the value for element 10: 4
Enter the value for element 11: 5
Enter the value for element 12: 6
Enter the value for element 20: 7
Enter the value for element 21: 8
Enter the value for element 22: 9

Original Matrice
1 2 3
4 5 6
7 8 9

Rotated Matrice
7 4 1
8 5 2
9 6 3

-----
Process exited after 10.38 seconds with return value 0
Press any key to continue . . .
```

Task 10

```
#include <stdio.h>

int main()
{
    int r, coef = 1;

    printf("Enter number of rows: ");
    scanf("%d", &r);

    for(int i=0;i<r;i++) {
        for(int j=1;j<=r-i;j++){
            printf(" ");
        }

        for(int k=0;k<=i;k++) {
```

```
    if(k==0 || i==0) {  
        coef=1;  
    } else {  
        coef=coef*(i-k+1)/k;  
    }  
    printf("%d ", coef);  
}  
printf("\n");  
}  
  
return 0;  
}
```

 C:\Users\Anas Ali Khan\Documents\PF Lab8 task10.exe

Enter number of rows: 5

```
  1  
 1 1  
1 2 1  
1 3 3 1  
1 4 6 4 1
```

Process exited after 2.395 seconds with return value 0
Press any key to continue . . .